

Decision Making And Hybrid-Based Trajectory Planning for Autonomous Vehicle In Various Highway Scenarios

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Abstract

With the rising interest in autonomous vehicles, the importance of algorithms that ensure safe and comfortable driving has come to the forefront. Correct decision-making in real-time while responding to continuously changing dynamic environments is a highly challenging problem. In particular, environments such as highways, where driving occurs on structured roads, require even more accurate and rapid decision-making due to the high speeds of both the ego and surrounding vehicles. In such scenarios, safety may be the top priority. However, only focusing on safety in driving is somewhat disconnected from the surrounding traffic environment. In other words, safety and driving performance are in a trade-off relationship. Therefore, it is crucial to control this relationship, followed by planning a trajectory that matches those decisions. This study proposes an algorithm that addresses it. Once, to improve real-time computational power Tactical Decision Making (TDM) and Trajectory Generator (TG) module work with a hierarchy. The TDM model select optimal decision based on Spatio-Temporal Safety Gaps by predicting the surrounding environment. And then, the TG module operates to execute that which is from TDM module. To incorporate the advantages of traditional path planning methods within TG module, a Hybrid Method is proposed. Additionally, this study aims to reduce computational cost by planning longitudinal and lateral trajectories separately. The two trajectories are generated as polynomial; the boundary conditions for the lateral direction are determined by lane-change decision, while for the longitudinal are set according to the surrounding traffic conditions. This approach is expected to allow efficient management of computational costs, leading to improved efficiency in rapidly changing environments.

Key Words: Tactical Decision Making (TDM), Trajectory Generator (TG), Hybrid - Based Method

1. Introduction

Autonomous vehicles(Avs) play an important role in transportation industry. Specifically, most of them spent much of their time in highway. Therefore, there are many research deal with Avs in this environment. Highway Scenario has several characteristics. First, it's a structured road and simple road regulation. The only thing that exists is a velocity limit. Unlike urban, there are no sudden pedestrian and traffic lights. Second, very fast changing scenes. Otherwise, traffic jams, mean velocity of surrounding vehicles is 80~110 km/h and faster than that. Considering these characteristics, it appears that decision-making must be somewhat conservative due to the significant safety risks involved. Additionally, rapid TDM and TG are necessary to keep pace with the rapidly

changing environment. Addressing these requirements will likely involve effectively using the simple road geometry and the characteristics of the highway situation. The longitudinal and lateral trajectories are generated using the Behavior Type created by TDM as the boundary condition. This hierarchical structure leads to a more efficient workflow. The TG module commonly utilizes two classical methods:

1) Sampling-Based Method

There is a random sampling approach that generates nodes without any constraints and a Deterministic Sampling Method where several constraints and search directions are predefined [4]. The former method has a low probability of generating the

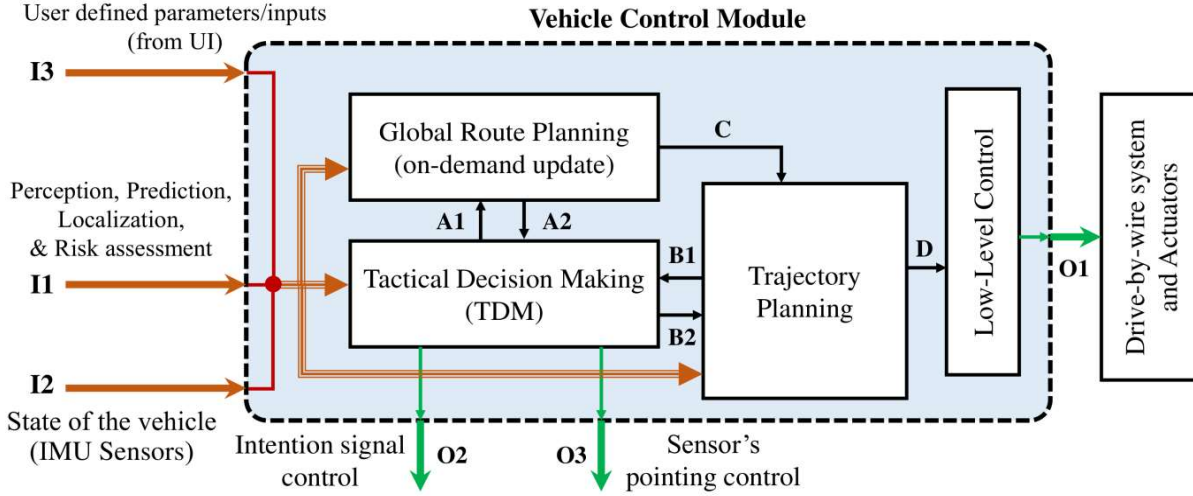


Fig. 1. The overall structure of the control module for Avs. we focus on TDM and Trajectory Planning module in this review.

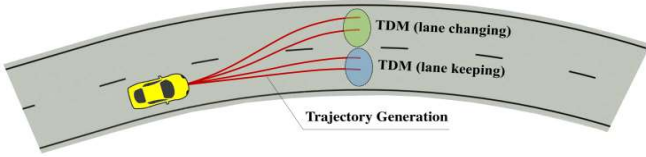


Fig. 2. Example illustration for sampling method for lane change and lane keeping maneuver

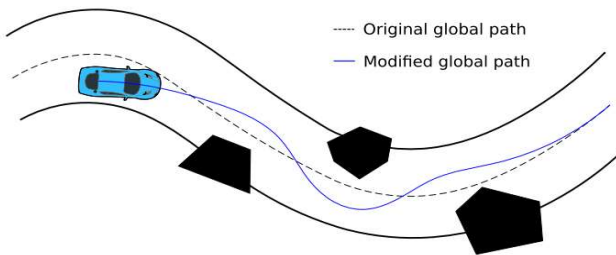


Fig. 3. Example illustration for optimization based trajectory in specific scene.

optimal trajectory, and as a result, the time cost is not negligible. Additionally, since the dynamics or kinematics of moving vehicles are not considered at all, there is a risk of infeasibility in the sampled trajectories. While Deterministic Sampling addresses the first drawback of the former method to some extent, the subsequent issue persists.

2) Optimization-Based Method

Unlike the sampling method, this approach allows for model-based planning and considers dynamic and kinematic constraints while algorithmically calculating the optimal path, thereby addressing feasibility issues and significantly increasing the likelihood of a global optimum. Moreover, there are methods such as MPC (Model Predictive Control), which can calculate not only the optimal guidance but also the control input which implements that policy. However, due to the numerous constraints and many iterations of complex algorithm, the computational cost is extremely high [4], making it unsuitable for real-time processing in highway.

3) Hybrid Method

Therefore, a Hybrid Method was implemented to incorporate the advantages of both approaches. To overcome the limitations of the constant velocity model in previous research [1] a boundary condition for longitudinal speed that reflects the traffic situation was generated. In this study, the controller is assumed to be ideal. So, the generated trajectory is accurately followed. Additionally, those method share a common weakness of not adequately responding to the uncertainty of the surrounding environment [4]. While this report does not address this issue, we consider developing methods to deal with it as part of our future work.

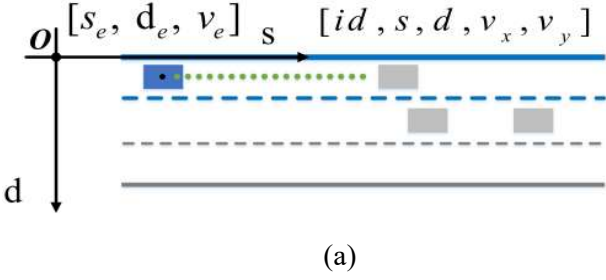


Fig.4. (a) Location information of ego vehicle in frenet frame

2. Methodology

2.1 Preview

Our system block diagram is in Fig.1 The sections were divided in the same order as the flow of the block diagram shown in Figure. Environment (Spatio-Temporal Safety Gap), Tactical Decision Making, Trajectory Generator and Infeasibility check such as collision with other vehicles.

2.2 Environment & Assumptions

a) Frenet States

For simplicity in trajectory modeling, we consider each direction independently by using Frenet frame as shown in Fig.4 (a). In this coordinate, ego vehicle is described by $[s, ds, dds, l, dl, ddl]$. s corresponds longitudinal direction and d lateral

offset from road center. So, overall planning task is implemented in this frame at relative inspection. Like Previous work [1],

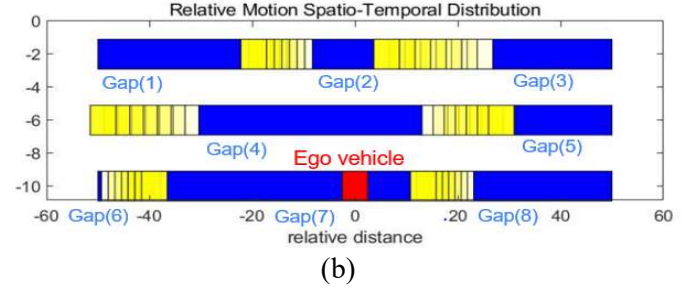
b) Spatio-Temporal Safety Gap

Surrounding vehicle's trajectories are predicted by constant velocity assumptions with duration time $T = 3s$. With prediction information and road network in presented coordinate, we can make deterministic scene which is called by Relative Motion Coordinate (RMC) [1]. And then, area without their trajectory is defined by Spatio-Temporal Safety Gap as shown in Fig.(5)

c) Most Important Object (MIO)

Because ego trajectory is received appropriate longitudinal boundary condition, we identify the most important (risky) object as shown in Fig (5). By prioritizing the surrounding things using specific algorithm. It is given by Fig (6).

In this way, the environment for TDM and TG is created.



Description of Spatio-Temporal Safety Gap red (ego-vehicle), yellow (surround vehicle)

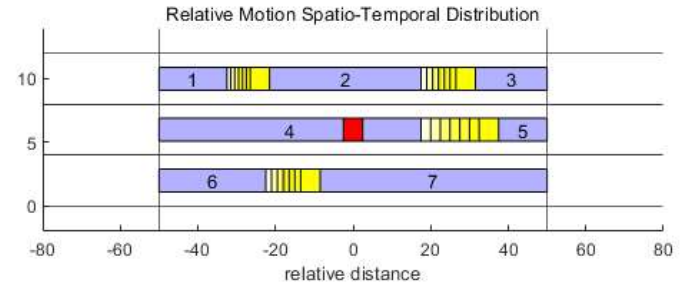


Fig. 5. Relative Motion Coordinate (RMC) and Safety Gap. Yellow is predicted trajectory of surrounding vehicle, Red is ego-vehicle. Blue is Safety Gap.

Algorithm 1: MIO (Most Important Object) Selection

Data: List of surrounding vehicles' longitudinal positions, Target Gap, Ego Lane
Result: MIO

```

1 Initialize MIOCandidates ← [];
2 RMC Boundary ← [-50, 50];
// Step 1: Select MIO Candidates within RMC Boundary
3 for each vehicle in surrounding vehicles do
4   if vehicle.longitudinalPosition within RMC Boundary then
5     MIOCandidates.append(vehicle);
6 if TargetGap.LaneNumber == EgoLane.LaneNumber then
// Step 2: Lane Keeping
7   MIO ← closest vehicle ahead in EgoLane from MIOCandidates;
8 else
// Step 3: Lane Change
9   MIOFront ← closest vehicle ahead in TargetLane from
    MIOCandidates;
10  MIOrear ← closest vehicle behind in TargetLane from
    MIOCandidates;
11  MIO ← (MIOFront, MIOrear);
12 return MIO

```

Fig. 6. Pseudo Code for finding MIO (Most Important Object)

d) Lane change vehicle

To compare the performance of our algorithm and related work [1], more complicated scenario is required like lane change scenario. The intentions of other are assumed to be pre-defined and there is no uncertainty either. However, there are many influences from this scene compared to linear moving. This topic will be covered in more detail later.

2.3 Tactical Decision Making (TDM)

a) Decision Topology

As mentioned in [1], connection between gaps as known as reachability is given by

$$Reach_{i,j} = \begin{cases} T, & |Gap_i.s - Gap_j.s| < \frac{Gap_i.l + Gap_j.l}{2} \\ F, & \text{otherwise} \end{cases} \quad (1)$$

From this, Topology is formed starting with the root node where the ego vehicle exists. This structure is used to calculate a specific cost efficiently. It will be discussed cost section.

b) Cost Function

To select optimal gap, total cost function and road cost function are given by

$$J^i = C_{road}^i + C_{traffic}^i + C_{arrival}^i \quad (2)$$

$$C_{road}^i = 1 - \frac{Gap_i.l}{50 \times 2} \quad (3)$$

Our relative frame boundary is $[-50, 50]$. So, total boundary is $50 \times 2 = 100$. Road cost represent degree of safety region. And then, traffic cost is made up traffic velocities and whether their position is front or rear from each Safety Gap. Equation is given by

$$C_{traffic}^i = C_{traffic,1}^i + C_{traffic,2}^i \quad (4)$$

$$C_{traffic,1}^i = \begin{cases} 0, & n_{car} = 0 \\ w_{car} \frac{Gap_{car}}{n_{car}}, & \text{otherwise} \end{cases} \quad (5)$$

$$C_{traffic,2}^i = \begin{cases} c_f \frac{vel_{ego}}{vel_{target}}, & s_{target} > Gap^i.l \\ c_b \frac{vel_{target}}{vel_{ego}}, & s_{target} < Gap^i.l \end{cases} \quad (6)$$

Compared to [1], we consider three more points. First, we analyzed the correlation with each member. Compared to other costs, the cost associated with vehicle numbers is highly sensitive regardless of degree of danger. Therefore, weight is multiplied to reduce that. Second, Governing equation reflects not only other's position but also speed. Third, by traffic coefficient, the risk is quantified. For example, the gap 10m away from the car in front is in a much more dangerous situation than the gap 30m away from the car in

front. The coefficient captures this difference well. Finally, Arrival cost function is given by

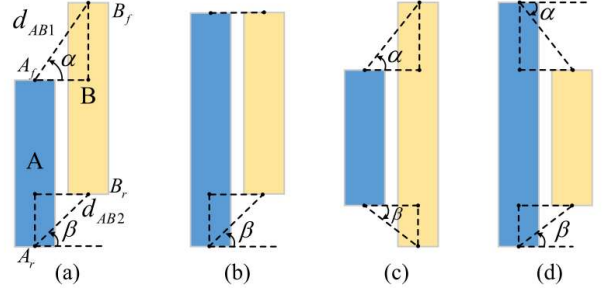


Fig.7. Four different scenes when calculating the arrival cost from A to B.

$$C_{AtoB}^i = C_{long}^i + C_{lat}^i \quad (7)$$

$$C_{long}^i = 1 - \frac{\min(A_f, B_f) - \max(A_r, B_r)}{Gap^{A,l}} \quad (8)$$

$$C_{lat}^i = \begin{cases} \frac{LaneWidth}{10}, & \Delta Idx_{lane} \leq 1 \\ 2.5, & \Delta Idx_{lane} > 1 \end{cases} \quad (9)$$

We add lateral cost to prevent TDM model to pick excessive lane change. Longitudinal cost describes difficulty of moving from A to B. In other words, the more overlapped region between them, the easier to move to B. So, Cost is low. And, by the topology structure, cost evaluation is implemented only for gaps with reachable to the root node. If not reachable, this gap has maximum arrival cost.

Through this procedure, total cost is calculated for each of them by linear combination with weight parameter. And the gap with the lowest cost is selected as the optimal gap as Fig (8).

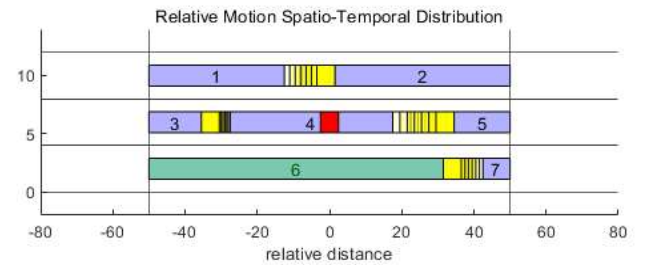


Fig. 8. Representation of RMC when TDM module has been implemented. Green is optimal gap.

2.4 Trajectory Generator (TG)

We select Hybrid scheme which sets cost each candidate. after sampling them using the boundary conditions.

a) Longitudinal Direction

To overcome limitation of constant speed model [1], it is important to select boundary condition appropriate for current scene. As mentioned above, Final MIO is picked by our algorithm and then its speed is to be boundary condition for trajectory planning. Safe Margin Sampling is used to verify sudden acceleration or deceleration. Therefore, Sampling number depends on the number of margin and duration time.

$$s(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + a_4 t^4 \quad (10)$$

$$\begin{cases} s(0) = s_0 \\ \dot{s}(0) = \dot{s}_0 \\ \ddot{s}(0) = \ddot{s}_0 \end{cases} \quad (11)$$

$$\dot{s}(T) = [v_{ego}, v_{ego} + \frac{1}{2}v_{rel}, v_{ego} + v_{rel}] \quad (12)$$

$$\ddot{s}(T) = \ddot{s}(T) \quad (13)$$

With eq (11) ~ (14), we can calculate polynomial coefficients using linear algebra like (16).

$$\begin{bmatrix} \dot{s}(T) - \dot{s}(0) - \ddot{s}(0)T \\ \ddot{s}(T) - \ddot{s}(0) \end{bmatrix} = \begin{bmatrix} 3T^2 & 4T^3 \\ 6T & 12T^2 \end{bmatrix} \begin{bmatrix} a_3 \\ a_4 \end{bmatrix} \quad (14)$$

b) Lateral Direction

Simple road conditions of highway and frenet framework facilitates sampling of lateral candidates. Because of one more boundary condition later direction, lateral model is given by

$$d(t) = b_0 + b_1 t + b_2 t^2 + b_3 t^3 + b_4 t^4 + b_5 t^5 \quad (15)$$

$$\begin{cases} d(0) = d_{ego} \\ \dot{d}(0) = \dot{d}_0 \\ \ddot{d}(0) = \ddot{d}_0 \end{cases} \quad (16)$$

$$\begin{cases} d(T) = d_{tergat} \\ \dot{d}(T) = \dot{d}(T) \\ \ddot{d}(T) = \ddot{d}(T) \end{cases} \quad (17)$$

As same as linear motion, coefficients are calculated by (18). In each decision term, candidate clusters are

generated with respect to duration time and boundary conditions. In this study, the former is 3 and the latter is [1s,2s,3s] In other words, the number of candidates is 9.

$$\begin{bmatrix} d(T) - d_0 - \dot{d}_0 T - \ddot{d}_0 T^2 \\ \dot{d}(T) - \dot{d}_0 - \ddot{d}_0 T \\ \ddot{d}(0) - \ddot{d}(T) \end{bmatrix} = \begin{bmatrix} T^3 & T^4 & T^5 \\ 3T^2 & 4T^3 & 5T^4 \\ 6T & 12T^2 & 20T^3 \end{bmatrix} \begin{bmatrix} a_3 \\ a_4 \\ a_5 \end{bmatrix} \quad (18)$$

c) Cost Function

To generate optimal trajectory, it is important to rank among them based on feasible way. Rather than complex optimization algorithm, we use candidates. Similar to TDM method, combination of cost function is applied. Through this method, it reduces the computational cost, and it is possible to select the most optimal trajectory among them. Formulation is given by

$$J^i = w_v C_v + w_a C_a + w_j C_j + w_k C_k + w_{ttc} C_{ttc} \quad (19)$$

$$C_v = \frac{1}{T} \int \frac{(v_d - \dot{s}(t))^2}{\Delta v_d^2} dt \quad (20)$$

$$C_a = \frac{1}{T} \int \frac{\ddot{s}(t)}{a_{max}^2} dt \quad (21)$$

$$C_j = \frac{1}{T} \int \frac{j_{long}(t)^2}{j_{max,long}^2} dt + \int \frac{j_{lat}(t)^2}{j_{max,lat}^2} dt \quad (22)$$

$$C_k = \frac{1}{T} \int \frac{\kappa(t)^2}{\kappa_{max}^2} dt \quad (23)$$

$$C_{ttc} = \frac{1}{T} \int \frac{(TTC_{max} - TTC(t))^2}{(TTC_{max} - TTC_{min})^2} dt \quad (24)$$

To allow the TG model to choose a variable velocity profile in some situations, TTC (Time To Collision) cost has been added to [1]. It is defined as shown in Fig (9).

We have some advantages of introducing it. Once, it is distinct from other indicator in safety perspective. For instance, in traffic jam scenario it is required to decelerate Avs. However, it is impossible to determine decrease velocity profile to optimal solution without TTC because convex formula of them make total cost to increase other than constant profile.

We have some advantages of introducing it. Once, it

Algorithm 2: TTC Calculation Algorithm

Data: Ego vehicle trajectory $\{s(t), d(t)\}$ and speed, MIO vehicle trajectory $\{s(t), d(t)\}$ and speed, TimeStep = 0.1s, MaxTTC, MinTTC

Result: Summation of TTC Cost at each TrajPoint

```

1 if Trajectory is Lane Keeping then
2   for each TrajPointIndex do
3      $s_{ego} \leftarrow s(t)$ ;
4      $s_{mio} \leftarrow mio.Trajlong(t)$ ;
5      $v_{ego} \leftarrow v(t)$ ;
6      $v_{mio} \leftarrow mio.vel$ ;
7     SafetyDistance  $\leftarrow s_{mio} - s_{ego}$ ;
8     RelativeSpeed  $\leftarrow v_{ego} - v_{mio}$ ;
9     if RelativeSpeed  $\leq 0$  then
10      Add 0 to TTC cost;
11   else
12     TTC  $\leftarrow$  SafetyDistance / RelativeSpeed;
13     if TTC  $\geq$  MaxTTC then
14       Add RelativeSpeed to TTC cost;
15     else
16       Add 0 to TTC cost;
17 else if Trajectory is Lane Change then
18   for each TrajPointIndex do
19      $s_{ego} \leftarrow s(t)$ ;
20      $d_{ego} \leftarrow d(t)$ ;
21     if Ego in initial lane then
22       MIO  $\leftarrow$  MIO in initial lane;
23     else
24       MIO  $\leftarrow$  MIO in change lane;
25     Repeat Calculation TTC for MIO;
```

Fig. 9. Pseudo Code for Calculate TTC Cost function

distinct from other indicator in safety perspective. For instance, in traffic jam scenario it is required to decelerate AVs. However, it is impossible to determine decrease velocity profile to optimal solution without TTC because convex formula of them makes total cost to increase other than constant profile. Next, this function, which considers the concept of time together rather than just distance, is more robust against dynamic environments that change as trajectories progress. Last, we can respond easily to vehicles that are faster than ego so that moving away. Thus, TG module not only samples various profiles but also picks one which is appropriate in this condition by TTC function.

d) Infeasibility check

We apply Non-Model-Based Sampling method when generating trajectory candidates. So, there is a risk to infeasibility which exceeds dynamic constraints (tire dynamics...) or collides with other objects. First risk is prevented from jerk, acceleration cost function. But second is never. We need to collision check each of them. Because having information of most risky object (MIO) already, we can check collide in identical time step. This algorithm is shown in Fig (10).

By selecting the collision trajectory in advance, the cost

Algorithm 3: Collision Detection for Trajectories

Input: Predicted MIO Trajectories $MIO(t)$, Sampled Ego Trajectories $Ego(t)$

Output: Collision status

```

1 for each trajectory in Ego(t) do
2   is_lane_change  $\leftarrow$  Check if the trajectory is a lane change;
3   if is_lane_change is False then
4     current_lane_MIO  $\leftarrow$  Find MIO trajectories in the current lane;
5     for each t in time_steps do
6       if Ego_Front(t)  $\cap$  MIO_Rear(t) then
7         Collision detected;
8         break;
9     end
10  end
11 else
12  current_lane_MIO  $\leftarrow$  Find MIO trajectories in the current lane;
13  target_lane_MIO  $\leftarrow$  Find MIO trajectories in the target lane;
14  for each t in time_steps do
15    if Ego.d(t) in current lane then
16      if Ego_Front(t)  $\cap$  MIO_Rear(t) then
17        Collision detected;
18        break;
19    end
20  end
21  else if Ego.d(t) in target lane then
22    for each MIO in target_lane_MIO do
23      if MIO is ahead of Ego then
24        if Ego_Front(t)  $\cap$  MIO_Rear(t) then
25          Collision detected;
26          break;
27        end
28      end
29    else if MIO is behind Ego then
30      if Ego_Rear(t)  $\cap$  MIO_Front(t) then
31        Collision detected;
32        break;
33      end
34    end
35  end
36 end
37 end
38 end
39 end
```

Fig. 10. Pseudo Code for Collision Detection each trajectory.

function calculation may also be simplified. Through this total process, which is from RMC to TG, final optimal trajectory is given. It is transmitted to AVs by changing to global coordinate system.

$$[s, ds, dds, l, dl, ddl] \rightarrow [x, y, vel, acc, \psi, \kappa] \quad (25)$$

3. Simulation results

3.1 Simulation setting

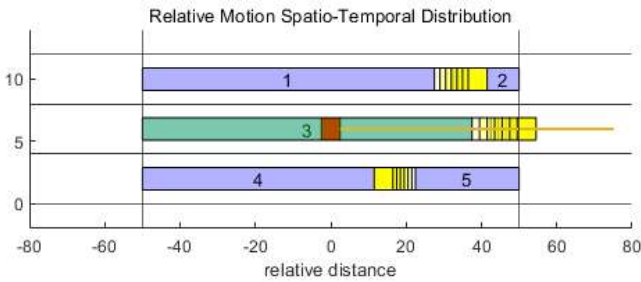
We separate linear motion and lane change motion so called “complicated scene” in surrounding vehicles viewpoint. Initial setting is as follows. $v_{ego,0} = 50\text{km/h} \sim 80\text{km/h}$, $a_{ego,0} = 0$. Surrounding has constant velocity assumption. In lane change scenario,

their trajectory is modelled by same manner as ego. Prediction time is 3s and RMC boundary is $[-50 \ 50]$. It is played MATLAB which is one scene and SIMULINK using Driving scenario Add-on which provides continuous scenes. During those scenarios, we checked velocity and acceleration profile, safety distance and preferred lane for performance verification of our algorithm. After proceeding first in various fixed scenes for weight parameter tuning, the continuous sequences are considered. The results in those scenes are compared to previous model [1] whose assumption is constant velocity. Dimensionless constants which are used to normalize numerical costs and facilitate the case study by obviously proposing each impact on each case. This constant value is given by [2], [3], [5], [6], [8], [9]. From those study, they are as follow. $a_{max} = 3m/s^2$, $v_{max} = 110km/h$, $v_{min} = 50km/h$, $j_{max,long} = 2.94m/s^3$, $j_{max,lat} = 0.9m/s^3$, $\kappa_{max} = 0.146 \frac{1}{m}$, $TTC_{max} = 5s$ and $TTC_{min} = 2.2s$

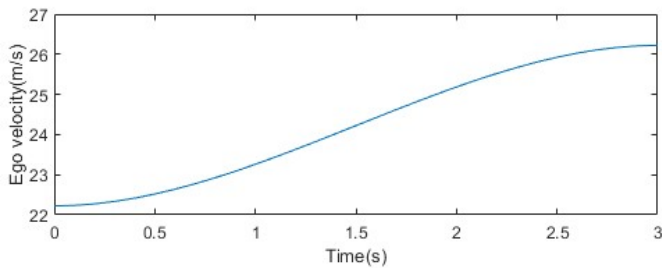
3.2 Following fast lead car

When surround traffic are faster than ego, the car has to accelerate to keep up with the flow. Accordingly, TDM module generates the boundary condition which is higher than ego and TG Module can select it well unless it's a dangerous route. Its design parameters are set up as follows to follow target velocity well than pay attention to stability.

$$\{w_v, w_a, w_j, w_k, w_{ttc}\}_{TG} = \{3, 1, 1, 3, 0.5\}$$



(a)



(b)

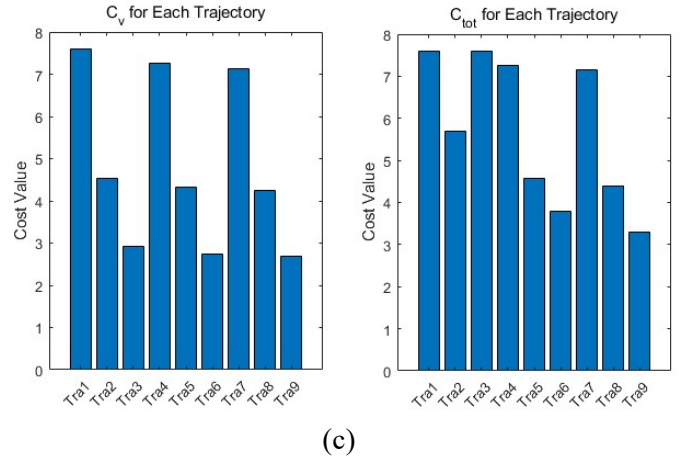
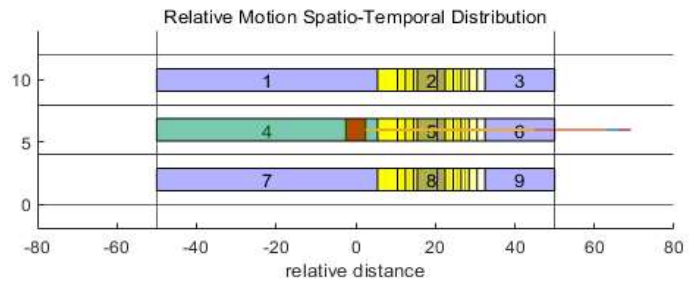


Fig. 11. (a) Visualization of result against the fast lead car scene. (b) Velocity profile from acceleration task. Optimal thing's duration time is 3s (c) Velocity & total cost each trajectory. Target tracking has major portion of total.

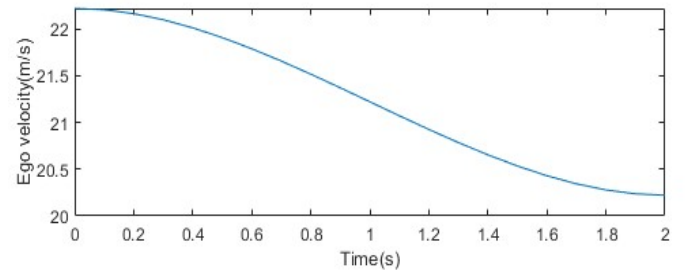
3.3 Keeping safety distance from lead car

When traffic jam, driving safety issues are highlighted. To keep safety distance above a certain level, ego must decrease speed. Thus, weight parameter which is related to safety is biggest around others when selection task is in progress. Fig (12). Since the environment requires stability mostly, safety-related weight is emphasized.

$$\{w_v, w_a, w_j, w_k, w_{ttc}\}_{TG} = \{1, 1, 1, 3, 0.5, 3.5\}$$



(a)



(b)

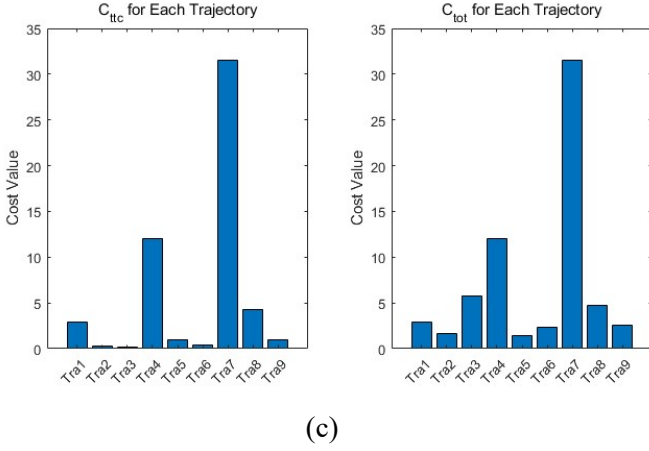


Fig.12. (a) Visualization of result against the slow lead car scene. (b) Velocity profile from deceleration task. Optimal thing's duration time is 2s. (c) TTC & total cost each trajectory. Constant trajectories TTC (1,4,7) is very high. So, they are picked never.

3.4 Lane Change

If there are lanes that are more safe or more comfortable to drive than current lane, AVs should go to that lane. He often drives to the highway exit or to follow a particular route. For this reason, lane change task is inevitable even the highway. Due to High-Order polynomial lateral trajectory model, overall paths are smooth as shown in Fig (13).

3.5 Complicated Scenario

The lane change situation of other vehicles falls into this category. The impact of changed environmental data is shown in the following Figure (14).

As can be seen, this scene first affects the Prediction part. Since the Prediction part is closely related to the find MIO block and make Safety Gap block, those also react sensitively. Consequently, blocks that rely on MIO, such as the generation of boundary conditions for the velocity profile, Collision Check, and TTC calculation, are sequentially affected. Likewise, TDM costs such as road cost and traffic cost that use traffic information directly are influenced too. This indicates a structure where changes in one block have a cascading impact on other parts of the system. Thanks to robustness of “Prediction Trajectory” and “Find MIO” algorithm this complicated scenario can be addressed. The result is shown below Fig(15).

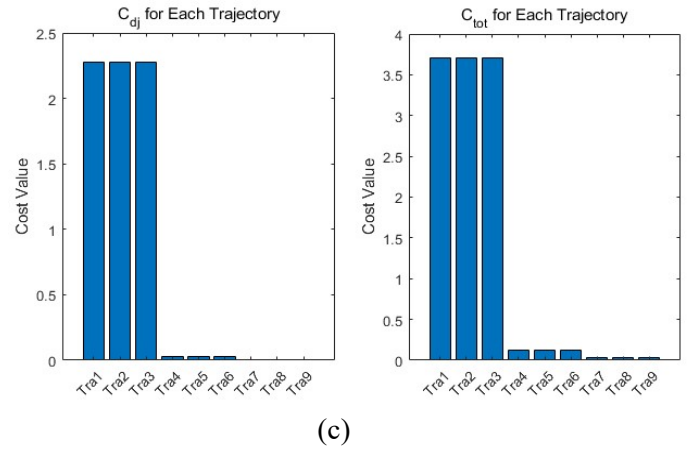
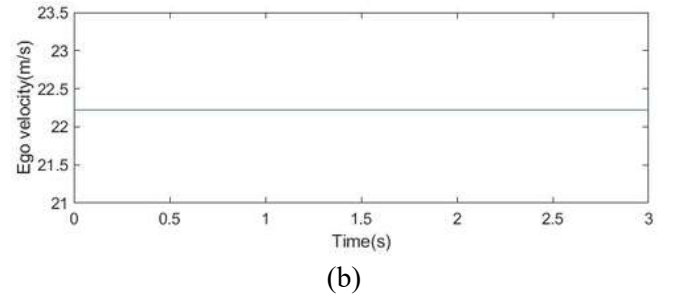
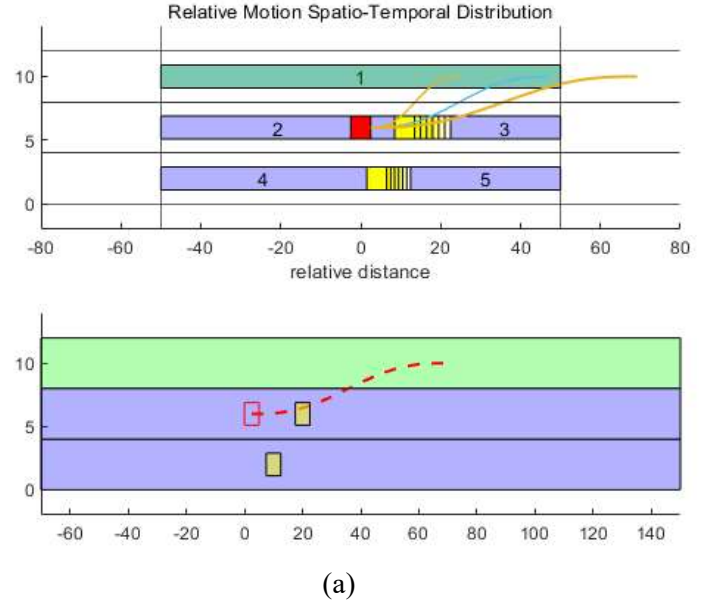


Fig. 13. (a) Expression of candidates and optimal lane change trajectory. (b) Constant velocity profile because there are no vehicles in the target lane. (c) In this scene, lateral jerk and curvature of path are dominant.

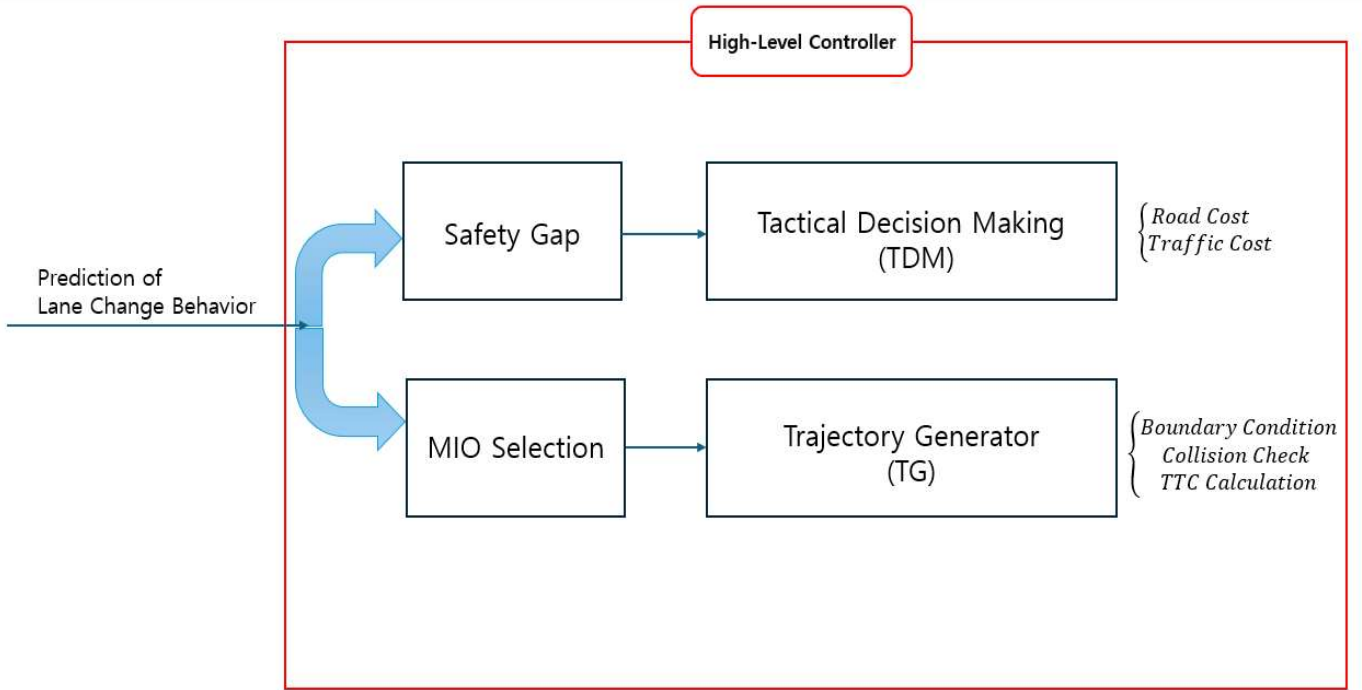


Fig. 14. Flow chart in algorithm from impact of other car's Lane Change Behavior.

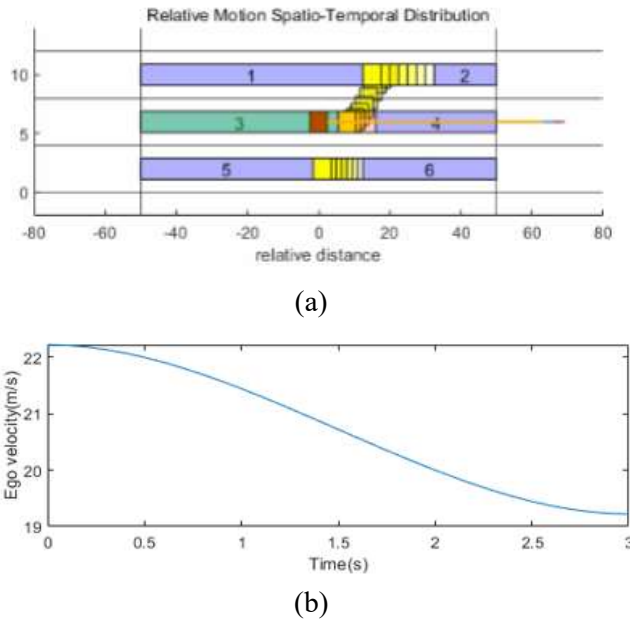


Fig. 15. (a) Visualization of result against the lane change of slow car scene. (b) Decelerate Velocity Profile.

3.6 Continuous simulations

We verify lastly this algorithm in the continuous environment using SIMULINK. Especially, to compare between it and previous thing [1], traffic jam scenario is added to original. Various profiles are represented in Fig. (17). The blue line can be seen broken during the process, indicating a case in which the simulation was stopped due to the detection of a collision between

another vehicle and an ego vehicle during the simulation. That point is collision point.

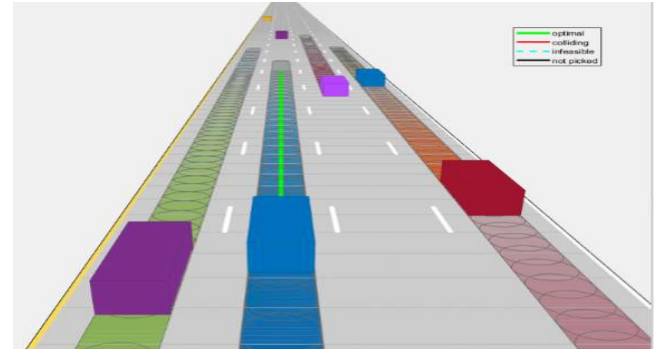


Fig. 16. Driving Scenario Add-on application.

4. Conclusion and Future work

From many simulation results, it is possible that this algorithm addresses various tasks from simple lane change to complicated acceleration or deceleration. In other words, modified TDM & TG module can implement not only one task which related to lane change but also the other related to target tracking even simultaneously. In future work, we improve TDM model to make it robust against from continuous environment by adding consistency in this module which is independent other costs similar to [7]. And then, in order to block from infeasibility completely, model-based planning such as Markov sequences which use dynamic model or optimization strategy like MPC (Model Predictive Control).

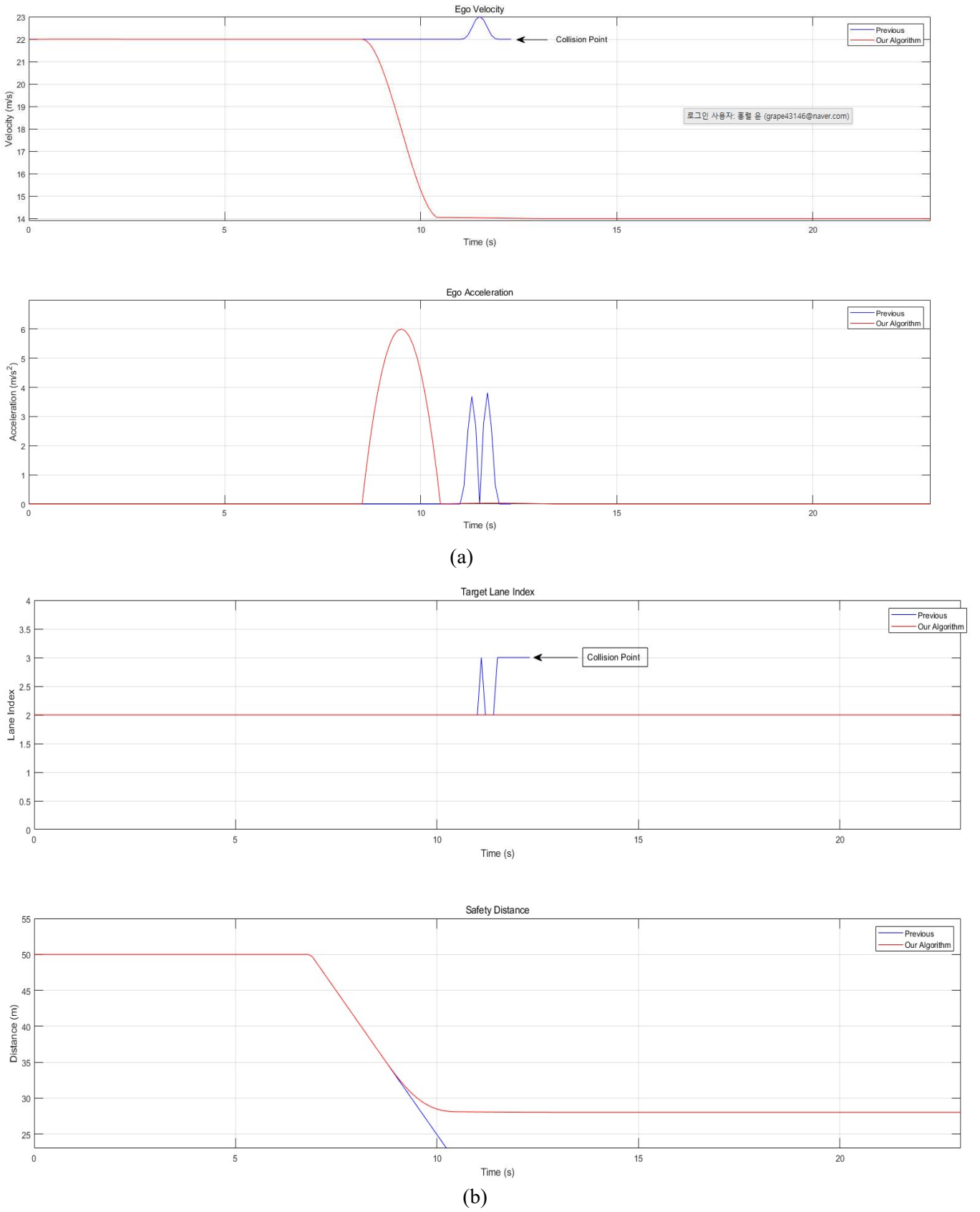


Fig. 17. Comparing Continuous Profile between previous and improved algorithm in traffic jam situation. Our Algorithm (red), Previous Algorithm (blue). (a) Velocity & Acceleration, (b) Target Lane & Safety distance.

5. Reference

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